1.1

p. 10 # 3, 5, 7, 13, 17, 19

3. $x_1 - x_2 = 1$
$x_1 + 2x_2 = 4$

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \xrightarrow{-r_1+r_2} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 3 & 3 \end{bmatrix} \xrightarrow{\frac{1}{3}r_2} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow{r_1+r_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

$x_1 = 2$
$x_2 = 1$

5. $$\begin{bmatrix} 1 & -4 & -3 & 0 & 7 \\ 0 & 1 & 4 & 0 & 6 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -5 \end{bmatrix}$$

-4 times r_3 plus r_2 replace r_2

3 times r_3 plus r_1 replace r_1

7. $$\begin{bmatrix} 1 & 7 & 3 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 7 & 3 & -4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

no solutions

13. $x_1 - 3x_3 = 8$
$2x_1 + 2x_2 + 9x_3 = 7$
$x_2 + 5x_3 = -2$

$$\begin{bmatrix} 1 & 0 & -3 & 8 \\ 2 & 2 & 9 & 7 \\ 0 & 1 & 5 & -2 \end{bmatrix} \xrightarrow{-2r_1+r_2} \begin{bmatrix} 1 & 0 & -3 & 8 \\ 0 & 2 & 15 & -9 \\ 0 & 1 & 5 & -2 \end{bmatrix}$$

$$\xrightarrow{r_2 \leftrightarrow r_3} \begin{bmatrix} 1 & 0 & -3 & 8 \\ 0 & 1 & 5 & -2 \\ 0 & 2 & 15 & -9 \end{bmatrix} \xrightarrow{-2r_2+r_3} \begin{bmatrix} 1 & 0 & -3 & 8 \\ 0 & 1 & 5 & -2 \\ 0 & 0 & 5 & -5 \end{bmatrix} \xrightarrow{\frac{1}{5}r_3} \begin{bmatrix} 1 & 0 & -3 & 8 \\ 0 & 1 & 5 & -2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\xrightarrow[-5r_3+r_2]{3r_3+r_1} \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$x_1 = 5 \quad x_2 = 3 \quad x_3 = -1$

17.

$2x_1 + 3x_2 = -1$
$6x_1 + 5x_2 = 0$
$2x_1 - 5x_2 = 7$

$$\begin{bmatrix} 2 & 3 & -1 \\ 6 & 5 & 0 \\ 2 & -5 & 7 \end{bmatrix} \overset{\begin{matrix} \\ -3r_1+r_2 \\ -r_1+r_3 \end{matrix}}{\sim} \begin{bmatrix} 2 & 3 & -1 \\ 0 & -4 & 3 \\ 0 & -8 & 8 \end{bmatrix} \sim \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & -1 \\ 0 & -4 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3/2 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{bmatrix} \quad 4r_2+r_3$$

inconsistent

19. $$\begin{bmatrix} 1 & h & 4 \\ 3 & 6 & 8 \end{bmatrix} \overset{-3r_1+r_2}{\sim} \begin{bmatrix} 1 & h & 4 \\ 0 & 6-3h & -4 \end{bmatrix}$$

$6 - 3h \neq 0$
$h \neq 2$

1.2

p. 21 # 3*, 7, 13, 19, 21

3. $\begin{bmatrix} \textcircled{1} & 2 & 4 & 8 \\ 2 & 4 & \textcircled{6} & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \sim \begin{matrix} \\ -2r_1+r_2 \\ -3r_1+r_3 \end{matrix} \begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & -3 & -12 \end{bmatrix} \sim \begin{matrix} \\ -\frac{1}{2}r_2 \\ -\frac{1}{3}r_3 \end{matrix} \begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{bmatrix}$

$\sim \begin{matrix} \\ \\ -r_2+r_3 \end{matrix} \begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} -4r_2+r_1 \\ \sim \end{matrix} \begin{bmatrix} \textcircled{1} & 2 & 0 & -8 \\ 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix} \sim \begin{matrix} \\ -3r_1+r_2 \end{matrix} \begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{bmatrix} \sim \begin{matrix} \\ -\frac{1}{5}r_2 \end{matrix} \begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & 1 & 3 \end{bmatrix}$

$\begin{matrix} -4r_2+r_1 \\ \sim \end{matrix} \begin{bmatrix} 1 & 3 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{bmatrix}$

$x_1 = -5 - 3x_2$

$x_3 = 3$

x_2 free

13. $\begin{bmatrix} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} r_1+r_3 \\ \sim \end{matrix} \begin{bmatrix} 1 & -3 & 0 & 0 & 9 & 2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\begin{matrix} 3r_2+r_1 \\ \sim \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 0 & -3 & 5 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$x_1 = 5 + 3x_5$

$x_2 = 1 + 4x_5$

$x_4 = 4 - 9x_5$

x_3 and x_5 free

19. $x_1 + hx_2 = 2$

$4x_1 + 8x_2 = k$

$\begin{bmatrix} 1 & h & 2 \\ 4 & 8 & k \end{bmatrix} \sim \begin{matrix} \\ -4r_1+r_2 \end{matrix} \begin{bmatrix} 1 & h & 2 \\ 0 & 8-4h & k-8 \end{bmatrix}$

$8 - 4h = 0 \quad h = 2$

$k - 8 = 0 \quad k = 8$

a) no solution

$h = 2, k \neq 8$

b) unique solution

$h \neq 2$

c) infinite solutions

$h = 2, k = 8$

21.

a. false (one reduced)
b. false
c. true
d. true
e. false (00005)

1.3

p. 32 #3*, 11, 13, 15, 19, 25

3. $u = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$v = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$

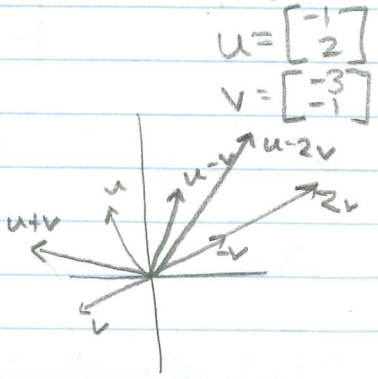
$-v = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$-2v = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

$u+v = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$

$u-v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$u-2v = \begin{bmatrix} 5 \\ -4 \end{bmatrix}$



NO? → 3rd ed OK

11. $\begin{bmatrix} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

yes

13. $\begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ -2 & 8 & -4 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 2 & 3 \\ 0 & 3 & 5 & -7 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ NO

15. $\begin{bmatrix} 1 & -5 & 3 \\ 3 & -8 & -5 \\ -1 & 2 & h \end{bmatrix} \sim \begin{bmatrix} 1 & -5 & 3 \\ 0 & 7 & -14 \\ 0 & -3 & 3+h \end{bmatrix} \sim \begin{bmatrix} 1 & -5 & 3 \\ 0 & 1 & -2 \\ 0 & -3 & 3+h \end{bmatrix} \sim \begin{bmatrix} 1 & -5 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & h-3 \end{bmatrix}$

$h-3=0 \rightarrow h=3$

19. $v_1 = \begin{bmatrix} 8 \\ 2 \\ -6 \end{bmatrix}$ $v_2 = \begin{bmatrix} 12 \\ 3 \\ -9 \end{bmatrix}$ $v_2 = \frac{3}{2} v_1$ ∴ all pts on line between 0 and v_1

→

25. $A = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 3 & -2 \\ -2 & 6 & 3 \end{bmatrix}$ $b = \begin{bmatrix} 4 \\ 1 \\ -4 \end{bmatrix}$

a) $b \notin \{a_1, a_2, a_3\}$, 3 vectors are

b) $W = \text{Span}\{a_1, a_2, a_3\}$

$$\begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 3 & -2 & 1 \\ -2 & 6 & 3 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 3 & -2 & 1 \\ 0 & 6 & -5 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 3 & -2 & 1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 1 & -2/3 & 1/3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

yes, ∞ vector in W
$b \in W$

∞??
free variables but still infinite?

$$\sim \begin{bmatrix} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

c) $a_1 = 1a_1 + 0a_2 + 0a_3$

1.4

p.40 #3*, 13, 15, 17, 21, 37, 39

3. $\begin{bmatrix} 1 & 2 \\ -3 & 1 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} = -2\begin{bmatrix} 1 \\ -3 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 2 \\ 1 \\ 6 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \\ -2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \\ 18 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 16 \end{bmatrix}$

$3\times 2 \quad 2\times 1 = \begin{bmatrix} 1(-2)+2(3) \\ -3(-2)+1(3) \\ 1(-2)+6(3) \end{bmatrix} = \begin{bmatrix} -2+6 \\ 6+3 \\ -2+18 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 16 \end{bmatrix}$

13. $\begin{bmatrix} 3 & -5 & 0 \\ -2 & 6 & 4 \\ 1 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 4 \\ 1 & -3 & -2 \\ 3 & -5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 4 \\ 0 & -4 & -6 \\ 0 & -8 & -12 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 1 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -6 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{bmatrix}$ yes

15. $\begin{bmatrix} 3 & -1 & b_1 \\ -9 & 3 & b_2 \end{bmatrix} \sim \begin{bmatrix} 3 & -1 & b_1 \\ 0 & 0 & b_2+3b_1 \end{bmatrix}$

Consistent: $b_1 = -\frac{1}{3}b_2$

pts w/ ↑ on line through origin

17. $\begin{bmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 2 & -1 & 4 \\ 0 & -4 & 2 & -8 \\ 0 & -6 & 3 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 2 & -1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 3 & 0 & 3 \\ 0 & 2 & -1 & 4 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

3 rows

No, not a pivot every row

21. No, 4 rows + 3 columns so not a pivot in every column

Span $\mathbb{R}^4$?

37. $\begin{bmatrix} 7 & 2 & -5 & 8 \\ -5 & -3 & 4 & -9 \\ 6 & 10 & -2 & 7 \\ -7 & 9 & 2 & 15 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -93/50 \\ 0 & 1 & 0 & 53/50 \\ 0 & 0 & 1 & -189/50 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ NO

39. $\begin{bmatrix} 10 & -7 & 1 & 4 & 6 \\ -8 & 4 & -6 & -10 & -3 \\ -7 & 11 & -5 & -1 & -8 \\ 3 & -1 & 10 & 12 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 & 10 \\ 0 & 0 & 1 & 0 & 10 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$ yes

1.5

p.47 #1,2,5,7,11,19

1. $Ax=0$ $\begin{bmatrix} 2 & -5 & 8 \\ -2 & -7 & 1 \\ 4 & 2 & 7 \end{bmatrix} \sim \begin{bmatrix} 2 & -5 & 8 \\ 0 & -12 & 9 \\ 0 & 12 & -9 \end{bmatrix} \sim \begin{bmatrix} 2 & -5 & 8 \\ 0 & -12 & 9 \\ 0 & 0 & 0 \end{bmatrix}$ yes, ∞ nontrivial b/c free var.

→ 2. $\begin{bmatrix} 1 & -2 & 3 \\ -2 & -3 & -4 \\ 2 & -4 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 3 \\ 0 & -7 & 2 \\ 0 & 0 & 3 \end{bmatrix}$ No free var ∴ no nontrivial

5. $\begin{bmatrix} 2 & 2 & 4 \\ -4 & -4 & -8 \\ 0 & -3 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $x_1 + x_3 = 0$, $x_2 + x_3 = 0$

$$\bar{x} = t\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

7. $\begin{bmatrix} 1 & 3 & -3 & 7 \\ 0 & 1 & -4 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 9 & -8 \\ 0 & 1 & -4 & 5 \end{bmatrix}$ $x_1 + 9x_3 - 8x_4 = 0$, $x_2 - 4x_3 + 5x_4 = 0$

$$\bar{x} = s\begin{bmatrix} -9 \\ 4 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} 8 \\ -5 \\ 0 \\ 1 \end{bmatrix}$$

→ 11. $\begin{bmatrix} 1 & -4 & -2 & 0 & 3 & -5 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 0 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

steps e.g.

$x_1 = 4x_2 - 5x_6$

$x_3 = x_6$

$x_5 = 4x_6$

$$x_2\begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_6\begin{bmatrix} -5 \\ 0 \\ 1 \\ 0 \\ 4 \\ 1 \end{bmatrix} + x_4\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

19. $a = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$ $b = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$ $x = a + tb$

$$\bar{x} = \begin{bmatrix} -2 \\ 0 \end{bmatrix} + t\begin{bmatrix} -5 \\ 3 \end{bmatrix}$$

1.6

p.54 #3, 11, 12, 13

3.

Input \ Output	F+P	M	S
F+P	.1	.1	.2
M	.8	.1	.4
S	.1	.8	.4

$p_F = .1p_F + .1p_M + .2p_S$

$p_M = .8p_F + .1p_M + .4p_S$

$p_S = .1p_F + .8p_M + .4p_S$

$$\begin{bmatrix} .9 & -.1 & -.2 & 0 \\ -.8 & .9 & -.4 & 0 \\ -.1 & -.8 & .6 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -22/73 \\ 0 & 1 & -52/73 \\ 0 & 0 & 0 \end{bmatrix}$$

$p_F = \frac{22}{73} p_S \qquad p_M = \frac{52}{73} p_S$

$p_F = \$30.14 \qquad p_M = \$71.23 \qquad p_S = \$100$

11. $MnS + As_2Cr_{10}O_{35} + H_2SO_4 \rightarrow HMnO_4 + AsH_3 + CrS_3O_{12} + H_2O$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7
Mn	1	0	0	1	0	0	0
S	1	0	1	0	0	3	0
As	0	2	0	0	1	0	0
Cr	0	10	0	0	0	1	0
O	0	35	4	4	3	12	1
H	0	0	2	1	0	0	2

$$\begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & -3 & 0 \\ 0 & 2 & 0 & 0 & -1 & 0 & 0 \\ 0 & 10 & 0 & 0 & 0 & -1 & 0 \\ 0 & 35 & 4 & -4 & 0 & -12 & -1 \\ 0 & 0 & 2 & -1 & -3 & 0 & -2 \end{bmatrix}$$

$x_1 = \frac{16}{327} x_7$

$x_2 = \frac{13}{327} x_7$

$x_3 = \frac{374}{327} x_7$

$x_4 = \frac{16}{327} x_7$

$x_5 = \frac{26}{327} x_7$

$x_6 = \frac{130}{327} x_7$

$16MnS + 13As_2Cr_{10}O_{35} + 374H_2SO_4 \rightarrow 16HMnO_4 + 26AsH_3 + 130CrS_3O_{12} + 327H_2O$

12. $\begin{bmatrix} 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 100 \\ 0 & 0 & 1 & -1 & -80 \\ 1 & 0 & 0 & 0 & 20 \end{bmatrix}$

$x_1 + x_4 = x_2$

$x_2 = x_3 + 100$

$x_3 + 80 = x_4$

$x_1 + 80 = 100$

x_4?

$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & 20 \\ 0 & 1 & 0 & -1 & 20 \\ 0 & 0 & 1 & -1 & -80 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$x_1 = 20$

$x_2 = x_4 + 20$

$x_3 = x_4 - 80$

$x_4 = x_4$

$x_4 > 80$

13. Same as notes

1.7

p. 60 # 1, 7, 9, 15, 17, 37

1. $\begin{bmatrix}5\\0\\0\end{bmatrix}, \begin{bmatrix}7\\2\\-6\end{bmatrix}, \begin{bmatrix}9\\4\\-8\end{bmatrix}$ $\begin{bmatrix}5&7&9\\0&2&4\\0&-6&-8\end{bmatrix} \sim \begin{bmatrix}5&7&9\\0&2&4\\0&0&4\end{bmatrix}$ ∴ unique solution and homogenous just has trivial ∴ lin. indep.

7. $\begin{bmatrix}1&4&-3&0\\-2&-7&5&1\\-4&-5&7&5\end{bmatrix}$ more columns then rows so not a pivot in every column at least 1 free variable $Ax=0$ has a non trivial ∴ lin. dep.

9. $\begin{bmatrix}1&-3&5\\-3&9&-7\\2&-6&h\end{bmatrix} \sim \begin{bmatrix}1&-3&5\\0&0&8\\0&0&h-10\end{bmatrix}$ a) $v \in \text{Span}(v_1, v_2)$ for no h b/c $0 \neq 8$

b) $\{v_1, v_2, v_3\}$ lin. dep $\forall h$

15. $\begin{bmatrix}5\\1\end{bmatrix} \begin{bmatrix}2\\8\end{bmatrix} \begin{bmatrix}1\\3\end{bmatrix} \cdot \begin{bmatrix}-1\\7\end{bmatrix}$

lin. dep b/c # vectors > # entries

17. $\begin{bmatrix}5\\-3\\-1\end{bmatrix} \begin{bmatrix}0\\0\\0\end{bmatrix} \begin{bmatrix}-7\\2\\4\end{bmatrix}$

lin. dep. b/c 0 vector

37. $v_1 \ldots v_4$ in $\mathbb{R}^4$ and $\{v_1, v_2, v_3\}$ lin. dep. then v_4 lin. dep.

T: v_4 either 0 vector or has 0 weight

1.8

p. 68 # 1*, 5, 7 (explain), 9, 11, 13, 19

1. $T(u)=\begin{bmatrix}2&0\\0&2\end{bmatrix}\begin{bmatrix}1\\-3\end{bmatrix}=\begin{bmatrix}2\\-6\end{bmatrix}$ $T(v)=\begin{bmatrix}2&0\\0&2\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix}=\begin{bmatrix}2a\\2b\end{bmatrix}$

5. $\begin{bmatrix}1&-5&-7&-2\\-3&7&5&-2\end{bmatrix}\sim\begin{bmatrix}1&-5&-7&-2\\0&-8&-16&-8\end{bmatrix}\sim\begin{bmatrix}1&-5&-7&-2\\0&1&2&1\end{bmatrix}$

$\sim\begin{bmatrix}1&0&3&3\\0&1&2&1\end{bmatrix}$ $x_1=3-3x_3$ $x_2=1-2x_3$ $\bar{x}=\begin{bmatrix}3\\1\\0\end{bmatrix}+t\begin{bmatrix}-3\\-2\\1\end{bmatrix}$

not unique ex: $\begin{bmatrix}0\\-1\\1\end{bmatrix}$

7. (6×5) $a=5$ # entries of x $b=6$ # entries of image

9. $\begin{bmatrix}1&-3&5&-5\\0&1&-3&5\\2&-4&4&-4\end{bmatrix}\sim\begin{bmatrix}1&-3&5&-5\\0&1&-3&5\\0&2&-6&6\end{bmatrix}\sim\begin{bmatrix}1&-3&5&-5\\0&1&-3&5\\0&1&-3&3\end{bmatrix}$

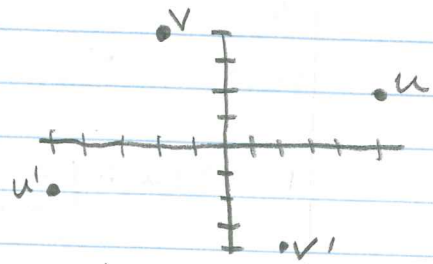
sol. to homo.

$\sim\begin{bmatrix}1&0&-4&10\\0&1&-3&5\\0&0&0&2\end{bmatrix}\sim\begin{bmatrix}1&0&-4&0\\0&1&-3&0\\0&0&0&1\end{bmatrix}$ $x_1=4x_3$ $x_2=3x_3$ $x_4=0$ $\bar{x}=\begin{bmatrix}4\\3\\1\\0\end{bmatrix}t$

11. $b=\begin{bmatrix}-1\\1\\0\end{bmatrix}$ $[A\ b]\sim\begin{bmatrix}1&-3&5&-5&-1\\0&1&-3&5&1\\0&0&0&-4&0\end{bmatrix}$ pivot each row so consistent. so b in range

13. $T(u) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} -5 \\ -2 \end{bmatrix}$

$T(v) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$



reflect over origin
rotate π radians

19. $A = \begin{bmatrix} 2 & -1 \\ 5 & 6 \end{bmatrix}$ $\quad T\left(\begin{bmatrix} 5 \\ -3 \end{bmatrix}\right) = \begin{bmatrix} 13 \\ 7 \end{bmatrix}$ $\quad T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 - x_2 \\ 5x_1 + 6x_2 \end{bmatrix}$

1.9

p. 78 # 1, 2, 3, 15, 17, 21, 25

1. $(4\times2)(2\times1)=(4\times1)$ $\begin{bmatrix} 3 & -5 \\ 1 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix}$

2. $(2\times3)(3\times1)=(2\times1)$ $\begin{bmatrix} 1 & -2 & 3 \\ 4 & 9 & -8 \end{bmatrix}$

3. $T(e_1)=e_1-3e_2$ $\quad T(e_2)=e_2$ $\quad \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$

15. $\begin{bmatrix} 2 & -4 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 3 \end{bmatrix}$

17. $T(x_1, x_2, x_3, x_4) = (x_1+2x_2, 0, 2x_2+x_4, x_2-x_4)$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

21. $\begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 3 \\ 8 \end{bmatrix}$ $\quad \begin{bmatrix} 1 & 1 & 3 \\ 4 & 5 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -4 \end{bmatrix}$

$x_1=7 \quad x_2=-4$

25. $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

free variable ∴ not one to one

not a pivot every row ∴ not onto

1.10

p. 86 #1, 5, 9

b) 1.

$$\begin{array}{c} \\ \text{Cal} \\ \text{P} \\ \text{Car} \\ \text{F} \end{array}\begin{bmatrix} \text{G} & \text{Q} \\ 110 & 130 \\ 4 & 3 \\ 20 & 18 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} G \\ Q \end{bmatrix} = \begin{bmatrix} 295 \\ 9 \\ 48 \\ 8 \end{bmatrix}\begin{array}{l} \text{cal} \\ \text{P} \\ \text{Car} \\ \text{F} \end{array}$$

a) X_1 [nutrients GM] + X_2 [nutrients Q] = [quant. nut.]

↳ amount GM ↳ amount Q

$$\begin{bmatrix} 110 & 130 & 295 \\ 4 & 3 & 9 \\ 20 & 18 & 48 \\ 2 & 5 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1.5 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

1.5 serv. GM

1 serv. Q

5.

$$\begin{bmatrix} 11 & -5 & 0 & 0 \\ -5 & 10 & -1 & 0 \\ 0 & -1 & 9 & -2 \\ 0 & 0 & -2 & 10 \end{bmatrix}$$

9.

$$\begin{bmatrix} .93 & .05 \\ .07 & .95 \end{bmatrix}\begin{bmatrix} 800000 \\ 50000 \end{bmatrix} = \begin{bmatrix} 769,000 \\ 531,000 \end{bmatrix}$$

$$\begin{bmatrix} .93 & .05 \\ .07 & .95 \end{bmatrix}\begin{bmatrix} 769000 \\ 531000 \end{bmatrix} = \begin{bmatrix} 741,720 \\ 558,280 \end{bmatrix}$$